1. An antique table increases in value according to the function $v(x) = 950(1.03)^x$ dollars, where x is the number of years after 1970.

a. How much was the table worth in 1970?
b. If the pattern indicated by the function remains valid, what was the value of the table in 1995?
c. Use a table or a graph to estimate the year when this table will reach double its 1970 value.

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a. The table was worth $ ______ in 1970.
(Round to the nearest cent as needed.)

b. The value of the table was $ ______ in 1995.
(Round to the nearest cent as needed.)

c. By the model, the value of this table reaches double its 1970 value in the year ______.

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*2. Suppose the quantity (in grams) of a radioactive substance present at time t is $Q(t) = 19{,}208\left(7^{-0.5t}\right)$, where t is measured in months.

a. How much will be present in 4 months?
b. How long will it take to reduce the substance to 8 grams?

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a. There will be ______ grams of the substance present in 4 months.
(Round to the nearest gram as needed.)

b. It will take ______ months to reduce the substance to 8 grams.
(Round to the nearest month as needed.)

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3. During a 10-year period of constant inflation, the value of a $300,000 property will increase according to the equation $v = 300{,}000e^{0.07t}$.
a. What will be the value of the property in 6 years?
b. Use a table or graph to estimate when this property will be worth double its original value.

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a. The value of the property in 6 years will be $ ______.
(Round to the nearest cent as needed.)

b. This property will be worth double its original value in ______ years.
(Round to the nearest year as needed.)